Distance-Constrained Dynamic Subtraction Games

Interval Structure, Certified Affine Regimes, and a Working Conjecture

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Abstract

We study an impartial take-away game where consecutive removals must differ by more than a fixed distance. We prove an interval classification of losing full-state rows and eventual P/N periodicity for fixed parameters. Using exact finite-window certificates and induction, we establish two finite collections of affine starting-position formulas: $42 \le k \le 53$ and $58 \le k \le 78$ for distance one. A general arithmetic obstruction explains why the $A = 54$ three-residue period degenerates at $k = 53$ and why no A-template can hold for $k \ge A$. We give counterexamples at adjacent parameters. A parameter-uniform explanation and a general hierarchy remain *Working on*. Each numerical claim is tied to a reproducible certificate; related literature is credited in the same article.

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1 Research context, question, and method

A move of size r after a previous removal s is legal only if $|r - s| > m$. This history dependence makes the state (n, s) essential. We ask which starting heap sizes are losing, why their binary sequence becomes periodic, and whether explicit periods and residues depend affinely on the maximum move k. Standard Sprague–Grundy theory motivates the P/N recurrence [1, 2, 3]. Prior work on repetition-free subtraction games provides an adjacent literature thread rather than a borrowed solution of the present $m = 1$ game [4].

We followed this process: (i) model the full state and derive an exact Boolean recurrence; (ii) independently cross-check it against naive P/N and direct Sprague–Grundy solvers; (iii) identify candidate affine residue patterns by exploratory scan; (iv) prove a finite-window implication by induction; (v) verify exact base windows and the necessary finite starting prefix for each fixed k; (vi) classify exceptional parameter boundaries, and (vii) identify conjectural claims explicitly as *Working on*. The source code and data are under `src/` and `data/certificates/`.

2 Theorem A: complete structural results

Proof status: Proved.

2.1 Rules, notation and outcome recurrence

Fix $k \geq 1$, $m \geq 0$ and a heap of $n \geq 0$ counters. From a state (n, s), where $s \in \{1, \ldots, k\}$ is the preceding removal, a legal move removes r counters such that $1 \leq r \leq \min(k, n)$ and $|r - s| > m$. The resulting state is $(n - r, r)$. With no previous removal, the first move has only the first restriction. Normal play applies: a player with no legal move loses. We call losing and winning states respectively $\mathcal{P}$- and $\mathcal{N}$-positions, following standard combinatorial-game terminology [1, 2, 3].

Define the P-state row and its potential P-reaching removals by

$$Q_n = \{s \in \{1, \ldots, k\} : (n, s) \text{ is } \mathcal{P}\}, \tag{1}$$

$$A_n = \{r : 1 \leq r \leq \min(k, n),\ r \in Q_{n-r}\}. \tag{2}$$

In particular $Q_0 = \{1, \ldots, k\}$, since there is no legal removal. Starting position n (no previous move) is a $\mathcal{P}$-position precisely when $A_n = \varnothing$.

Theorem 2.1 (Exact interval theorem)**.** *For every $n \geq 0$, the set Q_n is empty, full, or one interval of consecutive integers. More exactly, if $A_n = \varnothing$, then $Q_n = \{1, \ldots, k\}$. Otherwise*

$$\boxed{Q_n = [\max\{1, \max A_n - m\},\ \min\{k, \min A_n + m\}] \cap \mathbb{Z}.}$$

An interval whose lower endpoint exceeds its upper endpoint is empty. Any non-full row has at most $2m + 1$ entries.

Proof. The legal move r takes (n, s) to a losing state exactly when $r \in A_n$ and $|r - s| > m$. Hence (n, s) is losing precisely when *every* move in A_n fails its previous-removal constraint. Equivalently,

$|r - s| \le m$ for every $r \in A_n$. If A_n is empty, this holds for all $1 \le s \le k$. Otherwise the inequalities $r - m \le s \le r + m$ for all $r \in A_n$ intersect to $\max A_n - m \le s \le \min A_n + m$, clipped to $1 \le s \le k$. Its length is at most $2m + 1$. □

Lemma 2.2 (Finite-memory recurrence)**.** *For fixed k, m, Q_n is computed deterministically from $Q_{n-1}, \ldots, Q_{n-k}$ whenever $n \ge k$.*

Proof. The membership test $r \in Q_{n-r}$ determines A_n for each $1 \le r \le k$; then Theorem 2.1 determines Q_n. □

Theorem 2.3 (Eventual P/N periodicity)**.** *For every fixed $k \ge 1$ and $m \ge 0$, the sequence $(Q_n)_{n \ge 0}$ is eventually periodic, and so is the unrestricted starting-position P/N sequence.*

Proof. Each Q_n is a subset of $\{1, \ldots, k\}$, so there are at most 2^k row values. Consider consecutive ordered windows $(Q_n, \ldots, Q_{n+k-1})$. At most 2^{k^2} such windows exist. Theorem 2.2 makes the successor window a deterministic function of its predecessor once the recurrence has entered the full k-move range. Some two windows recur by pigeonhole, and all later windows agree by induction. Thus the rows are ultimately periodic. The starting $\mathcal{P}$ predicate $A_n = \varnothing$ depends on the preceding k rows for $n \ge k$, so it inherits eventual periodicity. □

Corollary 2.4 (Specialization $m = 1$)**.** *Every non-full Q_n has at most three consecutive elements.*

Scope. Eventual periodicity is not a statement about a least period or an affine formula in k. Those stronger claims require separate work.

3 Theorem B: a 21-parameter certified affine regime

Proof status: Proved by exact finite certificates plus induction (computer-assisted).

3.1 Certification lemmas

Lemma 3.1 (Window repetition)**.** *Fix k, m. Suppose for integers $p \ge 1, t \ge 0$, $Q_{t+i} = Q_{t+p+i}$ for every $0 \le i < k$. Then $Q_n = Q_{n+p}$ for all $n \ge t$.*

Proof. The given k equalities form the induction base. For $n \ge t + k$, each row used in the calculation of A_n is Q_{n-r} for $1 \le r \le k$, and the corresponding row for A_{n+p} is Q_{n+p-r}. These rows agree by induction; therefore their potential-removal sets and the resulting Q rows agree by the exact recurrence in Theorem 2.1. Since $n \ge t + k \ge k$, both states allow all removal sizes through k. □

Lemma 3.2 (Starting-position bridge)**.** *Under Lemma 3.1, the starting-position P/N sequence has period p for all $n \ge t + k$. A proposed p-periodic formula is valid for every $n \ge 0$ if it matches the exact starting P/N outcome for $0 \le n \le t + k + p - 1$.*

Proof. For $n \ge t + k$, each child row Q_{n-r} agrees with Q_{n+p-r}, and all removals $1 \le r \le k$ are available. Thus $A_n = A_{n+p}$, so the starting outcomes agree. The finite bound includes the entire earlier prefix and one complete period beyond $t + k$; any subsequent index reduces to that verified range. □

Theorem 3.3 (Exact certified $A = 86$ regime)**.** *For each integer* $58 \le k \le 78$*, the unrestricted starting position of* $D(k, 1)$ *with* n *counters is losing precisely when*

$$\boxed{n \bmod (2k + 88) \in \{0, 86, k + 87\}.}$$

The least positive period of this starting P/N sequence is $2k + 88$*.*

Finite computation and mathematical induction. Fix one of the 21 values $k = 58, \ldots, 78$. The script `src/verify_theorems_ab.py` computes each Q_n by the integer recurrence of Theorem 2.1. With $p = 2k + 88$, it finds the exact t shown in Table 1 and asserts the k row equalities in Lemma 3.1. It also verifies the proposed start predicate for every $0 \le n \le t + k + p - 1$. Lemmas 3.1 and 3.2 therefore prove the infinite-n claim for this fixed k. The script repeats these finite checks for each of the 21 listed integers. It additionally compares the rows and unrestricted starts with an independent direct Sprague–Grundy implementation over the verified range.

To establish minimality, set $R = \{0, 86, k + 87\}$ in the cyclic group of order p. If a nonzero shift fixes a set of three elements, its translation orbits must each have length three, so $R = \{0, p/3, 2p/3\}$. Its smaller nonzero element would therefore be $p/3$. Yet $p/3 = (2k + 88)/3 < 86$ for $58 \le k \le 78$, contradicting $\min(R \setminus \{0\}) = 86$. Thus the period is least. □

3.2 Exact reproduced certificates

Table 1: Theorem B, corrected finite-window certificates. The bridge endpoint is inclusive.

k	p	t	bridge end	independent SG
58	204	58	319	PASS
59	206	59	323	PASS
60	208	59	326	PASS
61	210	59	329	PASS
62	212	59	332	PASS
63	214	58	334	PASS
64	216	59	338	PASS
65	218	0	282	PASS
66	220	0	285	PASS
67	222	0	288	PASS
68	224	0	291	PASS
69	226	0	294	PASS
70	228	0	297	PASS
71	230	0	300	PASS
72	232	0	303	PASS
73	234	0	306	PASS
74	236	0	309	PASS
75	238	0	312	PASS
76	240	0	315	PASS
77	242	0	318	PASS
78	244	0	321	PASS

Unresolved extension (Working on). The 21 instances constitute a completed computer-assisted proof, not a uniform symbolic derivation for arbitrary k. Explicit P-row formulas explaining the numbers 86 and 88 and the boundary at $k = 78$ remain *Working on.* See Theorem C.

4 Theorem C: regime arithmetic, boundary obstructions, and a second certificate family

Full proposed parameter-hierarchy theorem status: Working on (not proved).

We investigate the empirical starting-position template

$$\mathcal{T}(k, A): \qquad P_0(n) \iff n \bmod p \in R, \qquad p = 2k + A + 2, \quad R = \{0, A, k + A + 1\}, \tag{3}$$

where k, A are positive integers and $P_0(n)$ means that the unrestricted start with n counters is losing. These observations concern this game's specific recurrence, not a claim previously established in the cited literature. The background on subtraction games and no-repetition variants motivates the question [3, 4].

4.1 Uniform analytic results: proved

Proposition 4.1 (Immediate upper bound on any valid template)**.** *If $\mathcal{T}(k, A)$ holds for $D(k, 1)$, then necessarily $k < A$.*

Proof. Equation (3) declares $n = A$ a losing unrestricted start, since $A \in R$. If $k \geq A$, however, removing exactly A counters is a legal *first* move from $n = A$ and leaves an empty heap, where the next player has no legal move. Therefore the first player wins from $n = A$, a contradiction. □

Proposition 4.2 (Arithmetic collapse of the three-residue template)**.** *Let $p = 2k + A + 2$ and $R = \{0, A, k + A + 1\} \pmod p$ with $k, A \geq 1$. The binary p-periodic indicator of R has a smaller positive period if and only if $k = A - 1$. In that case $p = 3A$, $R = \{0, A, 2A\}$, and the least positive period is A.*

Proof. The residues satisfy $0 < A < k + A + 1 < p$, so there are exactly three. If the indicator has a smaller period, a nonzero translation of the cyclic group $\mathbb{Z}/p\mathbb{Z}$ preserves R. Every orbit of a fixed translation has equal size, which must divide $|R| = 3$. Since a nonzero translation has no fixed point, its orbit size must equal three. The invariant set is therefore the orbit $\{0, p/3, 2p/3\}$. The smaller positive residue is A, so $A = p/3$, equivalently $3A = 2k + A + 2$, or $k = A - 1$. Conversely, if $k = A - 1$, then $p = 3A$ and $R = \{0, A, 2A\}$; membership depends only on divisibility by A, and A is clearly least because only residue zero is selected modulo A. □

For $A = 54$ the upper bound allows only $k \leq 53$, and the endpoint $k = 53$ forces an arithmetic period collapse. Both statements are unconditional *implications*; they do not by themselves establish that the template is realized by a game for any particular k.

4.2 Certified second affine family: proved with finite computation

Theorem 4.3 (Second fixed-parameter affine regime)**.** *For every fixed integer* $42 \le k \le 53$*, the unrestricted starting position in* $D(k, 1)$ *is losing exactly when*

$$\boxed{n \bmod (2k + 56) \in \{0, 54, k + 55\}.}$$

For $42 \le k \le 52$ *the least positive starting period is* $2k + 56$*. For* $k = 53$ *the least positive starting period is* 54*.*

Exact finite certificates and induction. For each of the twelve integers, the independent `src/verify_theorem_c.py` invokes the exact Theorem-A interval recurrence, seeks a repeated window of k P-state rows with displacement $p = 2k + 56$, and checks the formula through $t + k + p - 1$. The t and bridge endpoints are printed in Table 2. Theorem-B Lemmas 3.1 and 3.2 (reproved by identical elementary induction if this theorem is read independently) extend those finite observations to every n. Each row and starting-position outcome through the bridge is checked against a separate direct Sprague–Grundy solver. Thus the computation supplies explicit verifiable finite premises, while mathematical induction supplies the infinite consequence. Proposition 4.2 establishes the claimed least positive periods; the only collapse in the listed interval occurs at $k = 53 = A - 1$. □

4.3 Exact fixed-parameter certificate table

Table 2: Theorem C fixed-parameter certificates for $A = 54$.

k	candidate p	least	t	bridge end	SG
42	140	140	111	292	PASS
43	142	142	117	301	PASS
44	144	144	117	304	PASS
45	146	146	115	305	PASS
46	148	148	116	309	PASS
47	150	150	112	308	PASS
48	152	152	114	313	PASS
49	154	154	0	202	PASS
50	156	156	0	205	PASS
51	158	158	0	208	PASS
52	160	160	0	211	PASS
53	162	54	0	214	PASS

4.4 Nearby-parameter counterexamples: proved finite failures

A template that agrees with early rows need not repeat indefinitely. The exact recurrence and an independent direct P/N solver give the following first disagreement with the indicated A-template. Here "predicted" and "actual" refer to whether the unrestricted start is a P-position.

Table 3: Counterexamples to extending the fixed-A formulas.

k	A	first n	predicted P?	actual P?	direct P/N
41	54	42	no	yes	PASS
54	54	54	yes	no	PASS
55	54	54	yes	no	PASS
56	54	54	yes	no	PASS
57	54	54	yes	no	PASS
57	86	58	no	yes	PASS
79	86	492	yes	no	PASS
80	86	167	yes	no	PASS

Interpretation. The $k = 54, \ldots, 57$ failures of $A = 54$ at $n = 54$ follow already from Proposition 4.1. The $k = 79$, $A = 86$ failure at $n = 492$ is particularly instructive: several earlier blocks of starting outcomes happen to agree with the proposed formula, but its recurrence ultimately disagrees. One must not infer a theorem from a long numerical scan or a finite streak of residue matches.

4.5 What remains Working on

Conjecture 4.4 (Parameterized regime mechanism—Working on)**.** *There may exist a structured set of constants A and corresponding k-intervals for which $\mathcal{T}(k, A)$ is valid, with $A = 54$ and $A = 86$ representing two examples. An eventual symbolic classification may explain those constants, the actual parameter endpoints, and the complete P-state row intervals.*

Proof status: Working on (not yet proved). No assertion that an infinite hierarchy exists is proved. The finite certificates do not yield a parameter-uniform inductive formula for all rows. In particular we *have not* derived the numbers 54 and 86 from first principles, shown why the $A = 86$ regime terminates at $k = 78$, or ruled out isolated additional ranges of k with some other A.

4.6 Next verifiable proof obligations

The next step is to derive a finite, symbolic, piecewise affine formula for Q_n uniformly in k on a regime, prove each branch stable under the exact set recurrence, and justify the allowed parameter inequalities. A certificate-assisted theorem for finitely many k values must be distinguished from such a uniform proof.

5 Research conclusion and limitations

The interval theorem and general eventual periodicity are mathematical results independent of computation. Two exact finite-k starting-position classifications are established by computer-assisted certificates and induction, not merely by long observed sequences. Theorem C supplies a universal necessary condition $k < A$ for any valid A-template and identifies the exact period-collapse mechanism at $k = A - 1$. Its second family extends the previous data report to $k = 53$, where the least starting period is only 54 despite the larger displayed modulus 162. These results offer a concrete arithmetic boundary explanation, but not a uniform symbolic reason for the emergence of 54 and 86 or the

upper endpoint 78. *Working on:* a closed-form full-state classification, a parameter-uniform proof, and any infinite sequence of A values. Independent proof review and a broader priority literature review remain essential before a novelty claim.

Works cited and source verification

References

[1] R. Sprague, "Über mathematische Kampfspiele," *Tôhoku Mathematical Journal*, first series, vol. 41 (1935), pp. 438–444.
Original journal record: https://www.jstage.jst.go.jp/article/tmj1911/41/0/41_0_438/_article. The metadata and page span were checked against the journal listing.

[2] P. M. Grundy, "Mathematics and Games," *Eureka*, no. 2 (1939), pp. 6–8.
Bibliographical entry and link to scanned source at https://oeis.org/A002188/a002188.pdf; the bibliographical description is corroborated by OEIS. The scan was not used for any copied proof.

[3] E. R. Berlekamp, J. H. Conway, and R. K. Guy, *Winning Ways for Your Mathematical Plays*, vol. 1, second ed., A K Peters, 2001.
Publisher record: https://www.routledge.com/Winning-Ways-for-Your-Mathematical-Plays-Volume-1-2-Berlekamp-Conway-Guy/p/book/9781568811307; verified against publisher and author's publication listings.

[4] V. Gurvich, "On Repetition-free Subtraction Games and Vile-dopey Integers," RUTCOR Research Report RRR-01-2012, Rutgers University, January 2012.
Author's publication list: https://rutcor.rutgers.edu/~gurvich/publist.pdf (entry 240). A report transcription was located, but its detailed theorems were *not* invoked here.

Source-use boundary. The P/N and Sprague–Grundy methods are standard background and are credited above. All specific interval, finite-window, template-obstruction and finite-range arguments in these documents are written self-contained from the game definition, not borrowed from an external proof. Originality relative to *all* prior literature has not been established; a thorough review remains necessary.